VISWAPRIYAN R

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SUMMARY

Computer Science undergraduate and Freelance Full Stack Developer with experience developing client-focused web applications, interactive platforms, and data-driven solutions. Skilled in Python, Java, React.js, Django, FastAPI, and PostgreSQL, with hands-on experience in REST APIs, authentication, application workflows, performance optimization, and automated reporting. Also experienced in machine learning and computer vision, with a strong focus on building scalable and user-oriented software solutions.

TECHNICAL SKILLS

Programming Languages: Python, Java, C (Basics)
Frontend Development: HTML, CSS, JavaScript, React.js
Backend Development: Django, FastAPI, REST APIs, Authentication, Authorization
Databases: PostgreSQL, MySQL
Machine Learning & Computer Vision: PyTorch, YOLO, OpenCV, NumPy, Scikit-learn
Tools & Technologies: Git, GitHub, VS Code, Postman

EDUCATION

B.E. Computer Science and Engineering **CGPA: 7.98**
St. Joseph's College of Engineering, Chennai Graduating May 2028

HSSC (State Board) **88.16%**
Nellai Nadar Matriculation Higher Secondary School, Chennai Graduated Mar 2024

PROFESSIONAL EXPERIENCE

Kaspon Techworks Private Limited, Chennai **Jun - Jul 2026**
Full Stack Development Intern
- Completed a one-month internship focused on software development training using Python and Full Stack Development.

Big Bucks Innovation Private Limited, Chennai **Dec 2025**
Full Stack Developer Intern
- Gained hands-on AWS experience in deploying, managing, and optimizing scalable cloud applications and services.

FREELANCE EXPERIENCE

Ethyra Consulting Services - Official Website **ethyra.in**
Freelance Full Stack Developer
- Developed a responsive multi-page consultancy website featuring service pages, interactive content, and engagement experiences for CSR teams, NGOs, and development organisations.
- Engineered an interactive 3D storybook with realistic page-turn animations, optical loupe interactions, and keyboard, touch, and click navigation.
- Optimized navigation using speculative link prefetching and configured Cloudflare Pages caching for improved performance and asset delivery.

Rajmuni Civil Work Agency - Thermal & Seepage Inspection Portal
Freelance Full Stack Developer
- Developed a React-based inspection portal with separate Employee and Admin authentication workflows and a guided 7-step civil inspection form.
- Implemented employee submission synchronization, inspection management, and persistent browser-based storage for centralized administrative access.
- Built an interactive A4 report editor supporting thermal image comparisons, multi-page PDF generation, and editable Word document exports.

ETHYRA Impact - NGO Readiness & CSR Due Diligence Platform
Freelance Full Stack Developer
- Developed a full-stack NGO assessment platform evaluating 57 diagnostic criteria across 8 weighted compliance and governance pillars, with dynamic scoring and readiness classification.
- Built interactive analytics dashboards and a programmatic 8-page PDF report generator with compliance checklists and visualizations using Recharts and jsPDF.
- Implemented UPI payment submission and admin verification workflows with automated email notifications and PDF report delivery using Nodemailer.

PROJECTS

AI Football Detection & Match Analytics System
Python, PyTorch, YOLO, OpenCV, NumPy
- Built a computer vision pipeline using a custom-trained YOLO model (**97.6% mAP@50, 97.3% precision**) to detect players, referees, goalkeepers, and footballs in broadcast match videos.
- Implemented team classification using HSV masking and K-Means, ball possession tracking, IoU-based player tracking, and camera-motion-compensated sprint speed estimation.

AI-Powered Multi-Platform Marketing Intelligence & Campaign Optimization Platform
React.js, Python, FastAPI, PostgreSQL, Scikit-learn, Pandas, NumPy
- Developed a marketing platform for customer segmentation, campaign management, content generation, and performance analytics.
- Integrated machine learning models for bank campaign response prediction and RFM-based customer segmentation using K-Means clustering.
- Built campaign calendars and analytics dashboards to track social media performance, compare campaigns, and generate data-driven recommendations.

Citizen Science Data Aggregation Platform
Python, FastAPI, Scikit-learn, Google Gemini, LiveKit, SQLite, Scrapy, Next.js
- Developed a citizen science platform integrating machine learning-based anomaly detection using Scikit-learn and Great Expectations to validate environmental data.
- Implemented GraphRAG-powered contextual intelligence using LlamaIndex and Google Gemini, alongside LiveKit voice agents for interactive data reporting.
- Built geospatial dashboards and integrated Scrapy-based AQICN data scraping for cross-validation against official environmental data.

CERTIFICATIONS

- Meta Front-End Developer Professional Certificate (Coursera)
- Demystifying Network (NPTEL)
- Python for Data Science (NPTEL Elite)
- Database Management System (NPTEL Elite)
- Agentic AI Certified Foundations Associate (Oracle University)

Languages: English, Tamil, Telugu